

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I **M.Kasif Jalal (192424214)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Kasif Jalal

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet on demand.

Characteristics :

- On-demand self-service: Users can access resources whenever needed without human interaction.
- Broad network access: Services are available over the internet from different devices.
- Resource pooling: Resources are shared among multiple users using virtualization.
- Rapid elasticity: Resources can be increased or decreased quickly based on demand.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

- In the early days, organizations used mainframe computers, which were large, expensive, and shared by many users.
- The introduction of personal computers (PCs) made computing more affordable and available to individuals.
- As businesses grew, they started using data centers to store applications and information in one central location.
- The development of server virtualization allowed a single physical server to run multiple virtual machines, improving hardware efficiency and reducing costs.
- With the growth of the internet, software could be accessed online instead of being installed on individual computers.
- These advancements led to the development of modern cloud platforms like AWS, Microsoft Azure, and Google Cloud, which provide computing services on demand over the internet.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization :

- Server virtualization divides one physical server into multiple virtual machines.
- Each virtual machine works independently with its own operating system and applications.
- It improves hardware utilization by making better use of available resources.
- It reduces infrastructure costs and simplifies server management.
- It acts as the foundation of cloud computing by enabling efficient resource sharing.

Virtualization :

- Virtualization is a technology that allows one physical server to run multiple virtual machines.
- It improves hardware utilization by sharing resources efficiently.
- It can be used within an organization's local infrastructure without internet access.

- Its main purpose is to reduce hardware costs and improve resource management.

Cloud Computing :

- Cloud computing is the delivery of computing services such as servers, storage, software, and networking over the internet.
- It provides users with on-demand access to computing resources.
- It uses virtualization along with networking and automation to deliver services.
- Its main purpose is to provide scalable, flexible, and cost-effective computing services to users.

4. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

SOAP (Simple Object Access Protocol) :

- SOAP is a protocol used for exchanging information between applications.
- It uses XML as its data format.
- It provides high security and reliability for communication.
- It is suitable for banking, financial, and enterprise applications where security is important.
- It has higher communication overhead, making it slower than REST.

REST (Representational State Transfer) :

- REST is an architectural style used to build web services.
- It supports different data formats such as JSON, XML, and HTML, with JSON being the most commonly used.
- It is lightweight, simple, and faster than SOAP.
- It is widely used for cloud applications, web services, and mobile applications.
- It is more suitable for modern cloud environments because of its speed and flexibility.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

IaaS (Infrastructure as a Service) :

- Provides virtual servers, storage, and networking over the internet.
- Users can install and manage their own operating systems and applications.
- It gives organizations greater control over their infrastructure.
- Example: Amazon EC2.

PaaS (Platform as a Service) :

- Provides a platform for developing, testing, and deploying applications.
- Developers do not need to manage hardware or operating systems.

- It simplifies and speeds up the application development process.
- Example: Google App Engine.

SaaS (Software as a Service) :

- Delivers ready-to-use software through the internet.
- Users can access the software using a web browser without installing it.
- The service provider manages updates and maintenance.
- Example: Gmail.

XaaS (Anything as a Service) :

- Refers to the delivery of any IT service through the cloud.
- It includes services such as software, storage, databases, networking, and security.
- It provides flexibility by allowing users to access different services from a single cloud provider.
- Example: Microsoft 365.